# Google BoringSSL Name Constraints Bypass in SAN Validation Logic

**Google BoringSSL Name Constraints Bypass in SAN Validation Logic** - Timur Snoke, CERT/CC.

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## Content

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## Google BoringSSL Name Constraints Bypass in SAN Validation Logic

### Vulnerability Note VU#410595

Original Release Date: 2026-10-14 | Last Revised: 2026-10-14

### Overview

Five Name Constraints findings in BoringSSL (VU#410595.8–.12) describe cases where certificate names may not be constrained as intended. Records .8, .9, .10 and .12 concern the legacy crypto/x509 path; .11 concerns the newer pki path. Two have been fixed, one is accepted, one is disputed, and one was declined.

### Description

Name Constraints limit the names a subordinate certificate authority may issue. The reports concern how BoringSSL applies those limits to certain SAN (Subject Alternative Name) forms. The records do not share one status, and they do not all affect the same validation path.

**VU#410595.8** (**Fixed 2026-03-04**): A wildcard DNS name may not be treated as covering a specifically excluded host.

**VU#410595.9** (**Accepted**): A URI with a colon before `@` may cause the wrong host to be used when matching URI constraints.

**VU#410595.10** (**Disputed**): The finding concerns first-@ domain extraction in rfc822Name constraint matching. How consuming mail applications interpret the submitted multi-@ input, and whether that creates deployment exploitation, was not established.

**VU#410595.11** (**Rejected**): Trailing-dot handling in rfc822Name domains. The BoringSSL team declined this on 2026-02-28 as below its security threshold.

**VU#410595.12** (**Fixed 2026-03-04**): Trailing-dot DNS names were not always normalized on the legacy validation path.

Using BoringSSL in a product does not mean that product invokes an affected path or name form.

## Impact

These findings matter only where Name Constraints are enforced on an affected path, a trusted signature can be obtained over attacker-chosen SAN fields, and the accepted name is used for authorization. Mixed-library environments may accept or reject the same certificate differently. Presence of BoringSSL does not, by itself, mean that Chrome, Android, or other products are affected.

## Solution

BoringSSL fixes land in the upstream repository and reach products through each product's own update process.

- VU#410595.8 and .12 were fixed in commit 5774eca on 2026-03-04.
- VU#410595.9 is accepted and under coordination.
- VU#410595.10 is disputed.
- VU#410595.11 was declined.

CERT/CC is not coordinating BoringSSL fixes. For vendor status, contact the BoringSSL team through the Google Issue Tracker. Operators should confirm whether their deployment uses an affected path, apply available updates, and keep issued SANs well formed. A second validator helps only when it uses the same inputs and fails closed.

## Acknowledgements

Thanks to Oleh Konko for discovering and reporting these issues. This document was prepared with AI assistance by Timur Snoke.

## Vendor Information

410595

Filter by status: All Affected Not Affected Unknown

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Sort by: Status Alphabetical

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## References

- RFC 5280 §4.2.1.10 — Name Constraints
- RFC 1035 — Domain Names: Implementation and Specification
- RFC 3986 §3.2 — URI Authority Component
- BoringSSL Source Repository
- Google Issue Tracker
- BoringSSL commit 5774eca — wildcard and trailing-dot fixes

## Other Information

| **API URL: * | ***VINCE JSON*** | ***CSAF*** | | | ****Date Public:*** | *2026-10-14* | | | ***Date First Published:*** | *2026-10-14* | | | **Date Last Updated: * | 2026-10-14 15:11 UTC | | | *Document Revision: * | 2 | |

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